

SANTOSH SAHU

Hyderabad, Telangana, India | +91-9766064261 | santoshsk14320@gmail.com | linkedin.com/in/santosh-sahu | github.com/santoshsahu97660-arch

PROFESSIONAL SUMMARY

Aspiring Data Analyst / Python Developer (BCA 2025) with hands-on experience in end-to-end data workflow: dataset collection, cleaning, preprocessing, EDA, feature engineering, model building and evaluation. Built Computer Vision applications using OpenCV including real-time vehicle detection, tracking and traffic analysis. Skilled in SQL querying, reporting, visualization and debugging Python code. Comfortable with version control (Git) and writing reproducible notebooks.

TECHNICAL SKILLS

Languages	Python, SQL
Libraries	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, OpenCV
ML Techniques	Regression, Classification, Scaling, Cross Validation, Feature Engineering, Model Evaluation
Computer Vision	Video Capture, Grayscale, Gaussian Blur, Thresholding, Morphology, Contours, Haar Cascade
Database	MySQL (SELECT, WHERE, GROUP BY, ORDER BY, JOIN, Aggregations)
Tools	Jupyter Notebook, Google Colab, VS Code, GitHub, Excel (Pivot, Charts, VLOOKUP)

PROJECT EXPERIENCE

Real-Time Vehicle Detection & Tracking System (OpenCV)

- Captured frames using cv2.VideoCapture and processed frame-by-frame.
- Converted frames to grayscale using cv2.cvtColor for faster processing.
- Applied Gaussian blur (cv2.GaussianBlur) to reduce noise and smooth image.
- Performed thresholding and background subtraction for motion detection.
- Used cv2.findContours to identify moving objects and filtered by contour area.
- Tracked object centroid positions across frames to determine movement path.
- Drew bounding boxes using cv2.rectangle and displayed live feed via cv2.imshow.
- Optimized processing to maintain real-time FPS.
- Technologies: Python, OpenCV, NumPy.

Car Speed & Traffic Monitoring System

- Detected vehicles using Haar Cascade classifier.
- Computed displacement using frame differencing.
- Estimated approximate vehicle speed based on pixel movement and frame rate.
- Applied morphological operations to remove noise and false detection.
- Displayed tracking overlay on video output.
- Technologies: Python, OpenCV, NumPy.

Machine Learning Classification Model

- Loaded dataset using Pandas read_csv and handled missing values (fillna/dropna).
- Performed label encoding and feature scaling using StandardScaler.
- Split dataset using train_test_split.
- Trained models: Logistic Regression, KNN, Decision Tree, Random Forest.
- Evaluated using accuracy_score, confusion_matrix, precision and recall.
- Compared models and selected best performing algorithm.
- Visualized performance using Matplotlib graphs.

Sales Data Analysis & Visualization

- Cleaned dataset, removed duplicates and corrected datatypes.
- Used groupby and aggregation functions to analyze sales trends.
- Created charts (bar, line, histogram, heatmap) using Matplotlib and Seaborn.
- Identified top performing products and seasonal sales behavior.
- Prepared summary insights and reports.

PROFESSIONAL EXPERIENCE

- Computer Instructor – Moharsh Computer Education (Apr 2020 – Jun 2022).
- Taught computer fundamentals, MS-CIT and Tally ERP to 100+ students.
- Prepared practical assignments and maintained records using Excel.
- Explained technical topics and solved student technical issues.

EDUCATION

Bachelor of Computer Applications — Gondwana University — 2025

CERTIFICATIONS

- Python for Data Science
- SQL & Database Fundamentals
- Machine Learning Basics
- OpenCV Computer Vision Project Training

CORE COMPETENCIES

- Data Cleaning
- EDA
- Feature Engineering
- Model Evaluation
- Data Visualization
- SQL Querying
- Reporting

TOOLS

- Jupyter Notebook
- Google Colab
- VS Code
- GitHub
- Excel

LANGUAGES

- English
- Hindi

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge.

Place: Hyderabad

Date: _____

Santosh Sahu